



LESSON 5.4

THE TRIANGLE INEQUALITY THEOREM AND THE CONVERSE OF THE PYTHAGOREAN THEOREM

LESSON INTRODUCTION

MA.8.GR.1.3 Use the triangle inequality theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean theorem to determine if a right triangle can be formed from a given set of sides.

LEARNING TARGETS

- I can use the triangle inequality theorem to determine if a triangle can be formed from a given set of sides.
- I can use the converse of the Pythagorean theorem to determine if a right triangle can be formed from a given set of sides.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

ESSENTIAL QUESTIONS

- What side lengths does a triangle need to form a right triangle?
- What side lengths are needed to form a triangle?

LESSON NARRATIVE

In this lesson, students explore triangles. This lesson begins with students exploring the triangle inequality theorem to determine the relationship between the side lengths of triangles. Students build on this relationship to specifically explore right triangles and the relationship between the side lengths, relating this to the converse of the Pythagorean theorem.

COMPONENT SUMMARY

5.4.1 WARM-UP (~5 MIN.)

Notice and wonder about relationship between triangles and squares

5.4.3 GUIDED INSTRUCTION (10 MIN.)

Discover the converse of the Pythagorean theorem

5.4.5 YOUR TURN! (5 MIN.)

Determine if side lengths form a triangle or right triangle and error analysis

5.4.2 EXPLORATION (10 MIN.)

Discover the relationship between the side lengths of triangles

5.4.4 COLLABORATION (10 MIN.)

Discuss connection between the triangle inequality theorem and the converse of the Pythagorean theorem

5.4.6 WRAP-UP (~5 MIN.)

Reflection of understanding

RESOURCES

GLOSSARY

Key Terms:

- Triangle Inequality Theorem
- Converse of the Pythagorean theorem

Additional Terms:

- Right triangle

MATERIALS

- String, paper, or straws cut into five pieces from one to five inches
- (Optional) Ruler or tape measure

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may forget to square each number when applying the converse of the Pythagorean theorem.
- When students are squaring numbers, they may make the mistake of multiplying by two.

THINGS TO CONSIDER

- Encourage students to use the manipulatives throughout the lesson to help visualize the theorems.
- Consider providing students with more than the suggested five lengths of string, paper, or straws or providing different sets to different partners. Mixing the lengths would provide more information for the class.
- If time is an issue, then consider assigning 5.4.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

The 5.4.1 Warm-Up engages students in a Notice and Wonder routine. Encourage students to draw conclusions about the right triangles presented to prime them for the content they are diving into during this lesson.

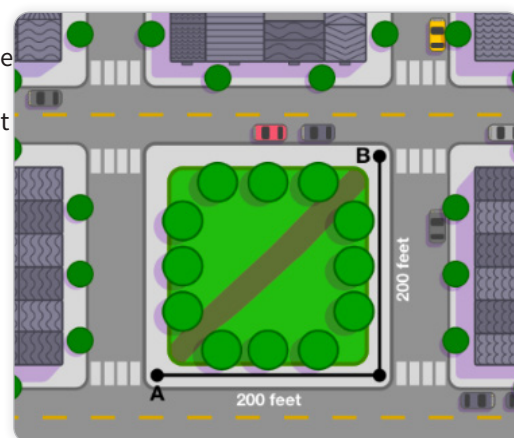
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

Encourage students to focus on their reasoning for their predictions. If students disagree, consider providing sentence stems to explore student reasoning and provide evidence to support their thinking. Using evidence will allow students to get to the correct answer and encourage active engagement.

DIFFERENTIATE INSTRUCTION

Consider providing Desmos Activity “Taco Truck” as support for remediating student conceptual understanding of the Pythagorean theorem and Converse of the Pythagorean Theorem. This digital activity provides students with an interactive, real-world context for understanding how to determine missing side lengths in right triangles and building understanding for why this is a useful skill.



5.4.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

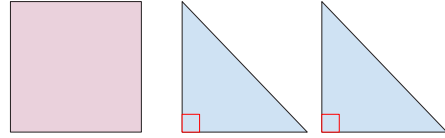
Students will discuss what they notice and wonder about the relationship between triangles and squares. Circulate as students make note of their observations and questions and highlight responses that note things like:

- The triangles are right triangles.
- I can cut the square diagonally to make two right triangles. I can combine the triangles to create a square.
- I wonder if I can find the area of the square using the triangles. I wonder if I can find the area of the triangles using the square.



5.4.1 Warm-Up: Notice and Wonder

1. A square and two right triangles are shown.



- a. What do you notice about the square and the two triangles?

Answers will vary. I notice if I put the two triangles together, I can make a square. I notice that if I cut a square diagonally, I can make two triangles.

- b. What do you wonder about the square and the two triangles?

Answers will vary. I wonder if I can find the area of a square using the area of a triangle. I wonder if I put the square and two triangles together if I could make a rectangle.



5.4.2 Exploration: Triangle Inequality Theorem

1. You will receive five different lengths of string, paper, or straw for this activity.
 - a. Attempt to make three different triangles on your desk using different sizes for the sides.

- b. Measure each side and record the lengths in the table. Indicate if the chosen side lengths formed a triangle or not.

Answers will vary. Side length are from 1 to 5.

Attempt	Side Length 1	Side Length 2	Side Length 3	Triangle Formed?
A	1	2	5	☐ Yes ☒ No
B	2	4	5	☒ Yes ☐ No
C	1	3	4	☒ Yes ☐ No

- c. For each attempt, compare the sum of two side lengths to the length of the third.
 - d. Record the results in the table.

Answers will vary.

	Attempt A	Attempt B	Attempt C
side 1 + side 2 > side 3	☐ True ☒ False	☒ True ☐ False	☐ True ☒ False
side 1 + side 3 > side 2	☒ True ☐ False	☒ True ☐ False	☐ True ☒ False
side 2 + side 3 > side 1	☒ True ☐ False	☒ True ☐ False	☐ True ☒ False

2. Complete the statements.
 - a. When the chosen side lengths formed a triangle, the side length comparisons were

☒ all true.
☐ all false.
☐ not all true.

- b. When the chosen side lengths did not form a triangle, the side length comparisons were

- ☐ all true.
☐ all false.
☒ not all true.



Triangle Inequality Theorem

The sum of the lengths of any two sides of a triangle is greater than the length of the third side.

3. Draw a picture to show what happens when the length of the third side of a triangle is longer than the sum of the other two sides.



5.4.2 EXPLORATION: TRIANGLE INEQUALITY THEOREM

TEACHER MOVES

Students will use the provided string, paper, or straws to test different combinations of sides to see if they will form a triangle. Provide each partner pair five different lengths of string, paper, or straws measuring one in., two in., three in., four in., and five in. Students may try to only pick side lengths that form a triangle. Encourage students to pick randomly to also have sides that do not form a triangle so that students can identify what side lengths can form a triangle and which cannot. Consider making sure students have at least one “no” answer in the first table.

TEACHER MOVES

Consider collecting information from the class and providing it for all students so that they have more information to answer question 2.

DIFFERENTIATE INSTRUCTION

Consider using this page for note-taking to encourage synthesis of the information. Auditory entry points can be provided if students discuss their answers before choosing their answers to allow time to process their thinking.

5.4.3 GUIDED INSTRUCTION: CONVERSE OF THE PYTHAGOREAN THEOREM

TEACHER MOVES

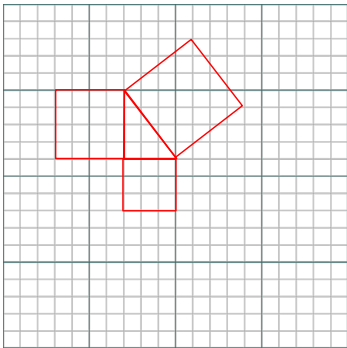
Students will discover how the squares of each leg can equal the squares of the hypotenuse. Students may need a review of how units work on a graph. If students struggle with drawing the square on the hypotenuse, suggest that the students use a ruler or a piece of paper to measure the length of the hypotenuse. Then the students can use the ruler’s edge or the paper’s corner as a right angle to draw the sides of the square.

TEACHER MOVES

Students are not using the Pythagorean theorem to solve but to verify if a triangle is a right triangle. The lesson combines the triangle inequality theorem and the converse of the Pythagorean theorem so that students understand that both are about the relationships of the sides of a triangle. Lessons 6 and 7 will build on the students’ understanding of the converse of the Pythagorean theorem to build student understanding of the Pythagorean theorem.

5.4.3 Guided Instruction: Converse of the Pythagorean Theorem

1. Use the grid to complete the steps.



- In the middle of the grid, draw a right triangle with legs 3 units long and 4 units long. Connect the legs to form the hypotenuse that is 5 units.
- Draw a square with side lengths of 3 units on the outside of the triangle and that shares a side with a leg of the triangle.
- Draw a square with side lengths of 4 units on the outside of the triangle and that shares a side with the other leg of the triangle.
- Draw a square with side lengths of 5 units on the outside of the triangle and that shares a side with the hypotenuse of the triangle.

2. Complete the table with the area of each square.

Side Length (units)	Area of Square (square units)
3	9
4	16
5	25

3. Complete the comparison for the sum of the areas of the squared legs to the square of the hypotenuse.
- $$3^2 + 4^2 = 5^2$$



5.4.4 Collaboration: The Triangle Inequality Theorem and the Converse of the Pythagorean Theorem

You will receive five different lengths of string, paper, or straw for this activity.

1. Discuss with your partner if a triangle can be formed with the side lengths of 2 centimeters (cm), 4 cm, and 5 cm.

- a. Test your prediction by attempting to form a triangle with the side lengths of 2 cm, 4 cm, and 5 cm.



- b. Discuss with your partner if the given lengths form a triangle. If the lengths formed a triangle, classify the type of triangle. If the lengths did not form a triangle, explain why not. Summarize your discussion.

Answers will vary. A triangle is formed because the sum of the lengths of any two sides is greater than the length of the third side. The triangle is not a right triangle. It appears to be obtuse.

- c. Complete the comparison for the sum of the areas of the squared legs to the square of the hypotenuse.

$$2^2 + 4^2 \neq 5^2$$

2. Discuss with your partner if a triangle can be formed with the side lengths of 6 cm, 8 cm, and 10 cm.

- a. Complete the comparison for the sum of the areas of the squared legs to the square of the hypotenuse.

$$6^2 + 8^2 = 10^2$$

- b. Based on the comparison, predict the classification of the triangle formed with side lengths of 6 cm, 8 cm, and 10 cm.

Right triangle

- c. Sketch a triangle with side lengths of 6 cm, 8 cm, and 10 cm.



- d. What is the classification the triangle formed?

Right triangle

- e. Discuss with your partner whether your prediction was correct or incorrect. Include how the comparison of the square of the legs to the square of the hypotenuse helped with your prediction. Summarize your discussion.

Answers will vary. I predicted it would be a right triangle because the sum of the squares of the legs are equal to the square of the hypotenuse.



The **Converse of the Pythagorean Theorem** states that if the lengths a , b , and c of the three sides of a triangle satisfy the relationship $a^2 + b^2 = c^2$ then the triangle is a right triangle.

5.4.4 COLLABORATION: THE TRIANGLE INEQUALITY THEOREM AND THE CONVERSE OF THE PYTHAGOREAN THEOREM

TEACHER MOVES

Students continue to use the materials from 5.4.2 Exploration.

DIFFERENTIATE INSTRUCTION

Using manipulatives such as string, paper, or straws will allow for a kinesthetic entry point to engage a variety of learners. Partner work allows for discussion and auditory entry points through mathematical discourse. Consider providing a visual for students to reference as they classify the types of triangles they are creating.

DIFFERENTIATE INSTRUCTION

As an extension, consider giving students side lengths that do not form a right triangle to discover what a nonexample should look like.

TEACHER MOVES

MA.K12.MTR.1.1: Participate in Effortful Learning

Encourage students to focus on their reasoning for their predictions. If students disagree, consider providing sentence stems to explore student reasoning and provide evidence to support their thinking. Using evidence will allow students to get to the correct answer.

TEACHER MOVES

Students discover this, so the theorem should make sense to them. There is no need at this point to further explain the Pythagorean theorem.

5.4.5 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider allowing students to graph the given lengths to help determine if they form a triangle or right triangle. Students can scale down the measurements and use one unit on the graph to represent one inch, foot, or yard. Creating a visual can help students access the material and determine whether the side lengths create a triangle/right triangle.

ENRICHMENT

Students may not understand why Gloria is correct or may not be able to accurately explain why she is correct. Ask guiding questions to encourage synthesis and help students verbalize their responses before putting them in writing.

- What is the formula we can use to determine if a triangle is a right triangle?
- Do the sides of this triangle satisfy the relationship when plugged into this formula? Why or why not?
- Can you classify this as a right triangle based on the converse of the Pythagorean theorem?

TEACHER MOVES

If students are unsure about what Pablo's statement tells them, refer them back to the Spotlight Tip on the Triangle Inequality Theorem found in the 5.4.2 Exploration.



5.4.5 Your Turn!

1. Determine if each set of side lengths form a triangle. If the side lengths form a triangle, determine if it is a right triangle.

Side Lengths	Triangle?	Right Triangle?
6 in., 12 in., and 13 in.	☐ Yes ☐ No	☐ Yes ☐ No
4 ft, 15 ft, and 17 ft	☐ Yes ☐ No	☐ Yes ☐ No
5 yd, 10 yd, and 15 yd	☐ Yes ☐ No	☐ Yes ☐ No

2. Pablo and Gloria were given the sides lengths 5 cm, 12 cm, and 17 cm then asked to determine if a right triangle was formed.

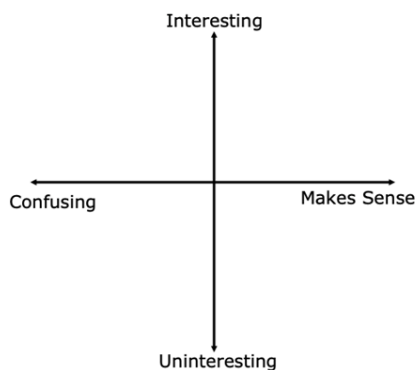
Pablo says it is a right triangle because $5 + 12 = 17$.	Gloria says it is not a right triangle because $5^2 + 12^2 \neq 17^2$.
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- a. Explain why Gloria is correct.
Gloria is correct because the Converse of the Pythagorean Theorem states that if the lengths, a , b , and c , of the three sides of a triangle satisfy the relationship $a^2 + b^2 = c^2$, then the triangle is a right triangle.
- b. What was Pablo's mistake or misconception?
Pablo did not square each side when comparing the sums of the legs and the hypotenuse.
- c. What does Pablo's statement tell you about the given side lengths?
The given side lengths do not form any triangle.

**5.4.6 Wrap-Up: Reflection**

1. Plot a point on the grid below to indicate your level of interest for today's learning target.

Answers will vary.

**5.4.6 WRAP-UP: REFLECTION****DIFFERENTIATE INSTRUCTION**

This tool allows an entry point for visual learners. Others may struggle with quantifying their reflection this way. Consider allowing these students to write their reflection in sentence format.